ITA03-MP Assignment 3

MOBILE COMPUTING

ASSIGNMENT 3 – CO3

Mode: Pair Work – 2 Students per Group
Duration: 1 Week
Total Marks: 50

Pair-Work Principle

Both students are responsible for understanding the complete assignment.

However, each question contains clearly identified individual responsibilities so that the contribution and understanding of each student can be independently verified.

Student A and Student B must not divide the assignment question-wise. Both students must participate in every question.

The individual responsibilities intentionally alternate across questions so that both students demonstrate evidence of:

Concept Understanding → Network Analysis → Protocol Design → Implementation/Simulation → Evaluation

Q1 – Wireless Network Architecture and Mobility Analysis 6 Marks

Part A – Introduction to Wireless Networks

Shared Group Work

Together:

- Understand the characteristics and components of wireless networks.
- Identify mobile nodes, wireless access points/base stations and communication links.
- Develop a suitable wireless network architecture.
- Identify bandwidth, connectivity and mobility constraints.

- analyse communication between mobile users and network infrastructure.
- Implement or simulate a small wireless communication scenario and test the communication flow.

Student A – Network Architecture Analysis

Student A must individually:

- Draw and represent the wireless network architecture.
- Identify the major network components and their roles.
- analyse wireless communication characteristics.
- Analyse the effect of limited bandwidth and wireless connectivity.
- analyse how mobile nodes communicate through the wireless infrastructure.

Student B – Mobility and Performance Analysis

Student B must individually:

- Design representative mobility scenarios.
- Analyse handoff and connectivity interruption.
- Study packet loss and transmission delay.
- Analyse the effect of changing network conditions.
- Compare communication behaviour under different mobility conditions.

Part B – Wireless Mobility Scenario

Shared Group Work

Together, develop and analyse a scenario in which a mobile user moves between wireless coverage areas.

Student A – Analytical Investigation

Student A must:

- Define successful communication and interrupted communication cases.

- Identify the sequence of mobility events.
- Record communication delay and packet delivery behaviour.
- Investigate increasing mobility conditions.
- Present the analytical evidence.

Student B – Mobility Logic

Student B must:

- Formulate the mobility decision sequence.
- analyse when a mobile node changes its point of attachment.
- Identify which communication conditions are affected.
- analyse the termination or completion condition for the scenario.
- Relate repeated mobility to communication efficiency.

Individual Verification

Faculty may ask Student A to analyse the wireless architecture and Student B to analyse the mobility scenario. Therefore, primary responsibility does not mean exclusive knowledge.

Q2 – Mobile IP and Mobility Management 7 Marks

Shared Group Work

The pair must:

- Study Mobile IP architecture and terminology.
- Identify Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- analyse agent discovery and registration.
- Analyse packet delivery to a mobile node.
- analyse tunnelling and mobility support.

- Conduct a small simulation or worked example of Mobile IP communication.

Student A – Mobile IP Operation

Student A is individually responsible for:

- analyseing the role of the Mobile Node.
- analyseing the role of the Home Agent.
- analyseing the role of the Foreign Agent.
- analyseing the role of the Correspondent Node.
- Describing registration and location updating.
- analyseing the packet-delivery and tunnelling process.

Student B – Mobile IP Challenge Analysis

Student B is individually responsible for:

- Identifying Mobile IP challenges.
- Analysing registration delay.
- Analysing tunnelling overhead.
- analyseing triangular routing.
- Analysing packet loss during mobility.
- interpreting security and scalability considerations.

Joint Design Question

Together answer:

If a mobile user frequently changes its network of attachment during an active communication session, what problems can occur in Mobile IP and how can they be reduced?

Evidence Expected

- Mobile IP architecture diagram.
- Registration/packet-flow sequence.
- Identification of mobility-related delays and overhead.
- Reasoned justification of the proposed improvement.

Cross-Understanding Requirement

Student A must be able to analyse the major Mobile IP challenges.

Student B must be able to analyse the complete Mobile IP registration and packet-delivery process.

Both students must be able to defend the answer.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks 7 Marks

Shared Group Work

Together:

- Study cellular wireless network architecture.
- Study ad hoc wireless network architecture.
- Identify infrastructure-based and infrastructure-less communication.
- Compare mobility and topology characteristics.
- Compare routing requirements.
- Develop a technical comparison diagram and table.

Student A – Cellular Network Investigation

Student A must individually:

- Represent cellular network architecture.
- analyse the role of base stations and infrastructure.
- Analyse centralized communication.
- analyse handoff between cells.

- Analyse scalability and coverage.
- Identify suitable cellular applications.

Student B – Ad Hoc Network Investigation

Student B must individually:

- Represent ad hoc network architecture.
- analyse communication between mobile nodes.
- Analyse dynamic topology.
- analyse multi-hop communication.
- Analyse the absence of fixed infrastructure.
- Identify suitable ad hoc applications.

Comparative Analysis

Parameter	**Cellular Network**	**Ad Hoc Network**
Infrastructure		
Communication		
Routing		
Mobility		
Topology		
Scalability		
Handoff		
Failure Handling		
Applications		

Joint Engineering Decision

Together determine:

For an emergency/disaster environment where fixed infrastructure may be unavailable, which network architecture would you select and why?

Cross-Defence

Student A must analyse why cellular networks depend on infrastructure.

Student B must analyse how ad hoc networks continue communication without fixed infrastructure.

Q4 – Applications of Mobile IP and Routing Requirements 7 Marks

Shared Group Work

Together:

- Identify real-world applications requiring mobility support.
- Select one application scenario such as emergency communication, mobile healthcare or intelligent transportation.
- Identify the mobile entities and communication requirements.
- Apply Mobile IP concepts to the selected scenario.
- Identify the routing requirements created by mobility.
- Evaluate the suitability of the proposed solution.

Student B – Application Architecture

Student B must individually:

- Select and describe an appropriate Mobile IP application.
- Identify mobile nodes and communication endpoints.
- Design the application network architecture.
- analyse how Mobile IP supports user mobility.
- Identify required network components.
- analyse the communication flow.

Student A – Performance and Requirement Analysis

Student A must individually:

- Analyse communication delay.
- Analyse packet delivery and connectivity.
- Identify mobility requirements.
- Analyse interruption during movement.
- Identify security and reliability requirements.
- Evaluate Mobile IP suitability.

Q5 – Proactive Routing Protocols in Mobile Networks 6 Marks

Shared Group Work

Together:

- Study proactive routing protocols.
- Understand continuous routing-table maintenance.
- Select a suitable protocol such as DSDV.
- Develop a mobile-node topology.
- Analyse routing-table updates.
- Evaluate routing behaviour under mobility.

Student A – Protocol Design

Student A must individually:

- analyse the principle of proactive routing.
- analyse DSDV operation.
- analyse routing-table maintenance.
- analyse sequence numbers and route updates.

- Draw a sample routing topology.
- Demonstrate route information before and after a topology change.

Student B – Performance Investigation

Student B must individually:

- Analyse routing-control overhead.
- Analyse route availability.
- Study convergence behaviour.
- Analyse control-message transmission.
- Compare behaviour for different network sizes.
- Evaluate delay and packet delivery.

Q6 – Reactive Routing and Protocol Comparison 7 Marks

Student A – Reactive Routing Investigation

Student A takes primary responsibility for reactive routing.

- analyse the concept of reactive routing.
- Study AODV.
- analyse Route Request (RREQ).
- analyse Route Reply (RREP).
- analyse Route Error (RERR).
- Analyse route discovery and route maintenance.
- analyse the effect of mobility on route discovery.

Student B – Comparative Verification

Student B must provide a comparative analysis of AODV and DSDV.

- Compare route discovery mechanisms.

- Compare routing-table requirements.
- Analyse control overhead.
- Compare initial communication delay.
- Analyse mobility handling.
- Identify suitable application scenarios.
- Justify the selection of a protocol for a given network condition.

Joint Question

For a highly dynamic mobile network with frequent topology changes, would you select DSDV or AODV? Justify your decision using routing overhead, delay and mobility considerations.

DAY-4 DESIGN CHALLENGE

Q7 – Design a Mobile Emergency Communication Network 10 Marks

The Day-4 problem must be solved collaboratively.

Neither student may work on this question independently and simply combine the answers later.

Scenario

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move continuously across different geographical areas. Fixed communication infrastructure may become unavailable, while mobile users must exchange critical information with minimum delay.

Stage 1 – Independent Design Proposal

Before interpreting the solution with each other:

Student A

Independently propose:

- A suitable wireless network architecture.

- A mobility-management strategy.
- The role of Mobile IP.
- A possible communication model.

Student B

Independently propose:

- A suitable routing strategy.
- Proactive/reactive protocol selection.
- Route maintenance approach.
- Failure-handling mechanism.
- Performance parameters.

Each student's initial proposal must be retained in the submission.

Stage 2 – Compare the Two Designs

The pair must compare where the designs agree, where they differ, strengths and weaknesses, Mobile IP requirements, routing suitability, expected overhead, expected delay and handling of node mobility.

Stage 3 – Joint Final Design

Using the strongest ideas from the interpretion, jointly develop the final mobile emergency communication solution.

- Network architecture showing mobile nodes, wireless links and routing infrastructure.
- Mobile IP components: Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- Registration and tunnelling process.
- Selected routing protocol and justification.
- Failure handling for node/link failure and network disconnection.

- Performance evaluation using delay, packet delivery and routing overhead.

Stage 4 – Divide the Validation Responsibilities

Student A – Analytical Validation

- Analyse Mobile IP overhead.
- Analyse mobility-related delay.
- Evaluate packet delivery.
- Analyse communication interruption.
- Justify the selected mobility-management mechanism.

Student B – Experimental Validation

- Design network test cases.
- Test different numbers of mobile nodes.
- Test different mobility conditions.
- Compare proactive and reactive routing behaviour.
- Record delay and packet-delivery results.
- Present experimental evidence.

Stage 5 – Cross-Defence

Student A must analyse: Why was the selected routing protocol appropriate for the emergency mobile network?

Student B must analyse: How does Mobile IP maintain communication when a mobile node changes its network location?

Stage 6 – Joint Evaluation

- Why the proposed network architecture is suitable.
- Why Mobile IP is required or not required.
- Why the selected routing protocol is appropriate.
- What network overhead is introduced.

- How mobility affects performance.
- What happens when connectivity is lost.
- Under what conditions the proposed solution may not be appropriate.

Conclude: What characteristics should a network designer analyse before selecting Mobile IP and a routing protocol for a highly mobile wireless environment?

CONTRIBUTION RECORD

Question	**Student A – Primary Contribution**	**Student B – Primary Contribution**	**Joint Work**
Q1	Network Architecture	Mobility Analysis	Wireless Network Analysis
Q2	Mobile IP Operation	Mobile IP Challenges	Mobile IP Architecture
Q3	Cellular Network	Ad Hoc Network	Comparative Analysis
Q4	Performance Analysis	Application Architecture	Application Evaluation
Q5	Protocol Design	Performance Investigation	Proactive Routing
Q6	Reactive Routing	DSDV–AODV Comparison	Routing Evaluation
Q7	Analytical Validation	Experimental Validation	Final Network Design

INDIVIDUAL VERIFICATION

Faculty may select any question, including a component for which the student was not designated as the primary contributor.

- analyse wireless network architecture.
- analyse Mobile IP operation and registration.
- analyse mobility challenges.
- Compare cellular and ad hoc wireless networks.
- analyse applications of Mobile IP.
- analyse proactive routing.
- analyse reactive routing.
- Compare DSDV and AODV.
- Justify protocol selection.
- Interpret experimental results.
- Justify network design decisions.
- Respond to a modified network condition.

A student should not answer: "My partner did this section."

Understanding of the complete submitted work is a requirement of the assignment.

NOTE: RECOMMENDED FACULTY VERIFICATION MODEL (VIVA)

For each student, conduct a brief 4–5 minute verification.

Question 1 – Primary responsibility

Ask one technical question from the component for which the student had primary responsibility.

Question 2 – Partner's responsibility

Ask one question from a component primarily completed by the partner.

Question 3 – Transfer question

Modify a condition and ask the student to reason about the consequence.

- If a mobile node moves from one network to another during an active session, what happens to its IP connectivity and how does Mobile IP address the problem?
- If the number of mobile nodes increases significantly, how would AODV and DSDV differ in routing overhead?
- If the network topology changes very frequently, which routing approach would you select and why?
- If the fixed infrastructure becomes unavailable during a disaster, how would an ad hoc network support communication?

This verification establishes individual attainment without requiring a second complete individual assignment.